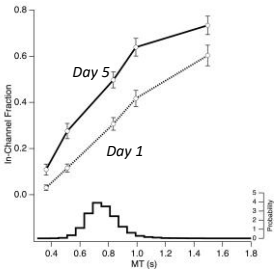
Neuro-pathologies of the motor system and aging

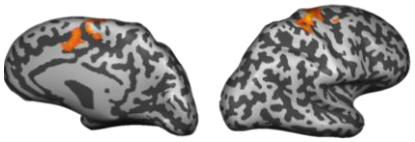
Lior Shmuelof
IEM
32 208
shmuelof@bgu.ac.il

Brain and Action Lab

מיומנות

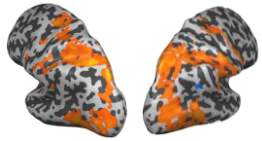


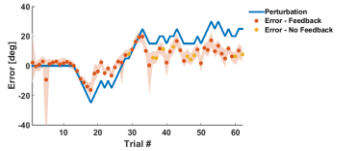




אדפטציה



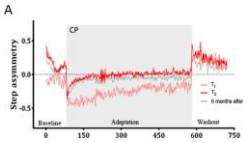




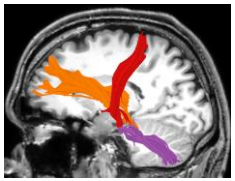
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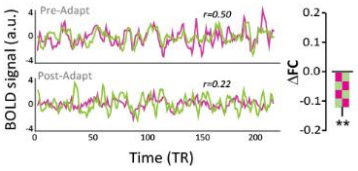


שיווי משקל

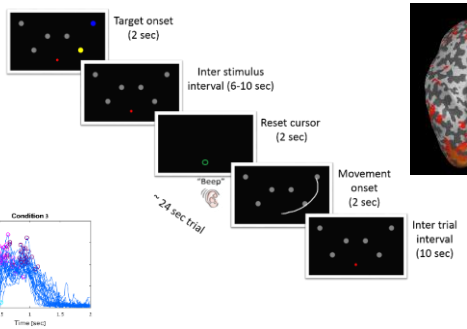


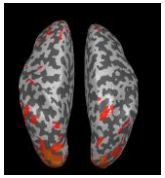


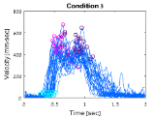




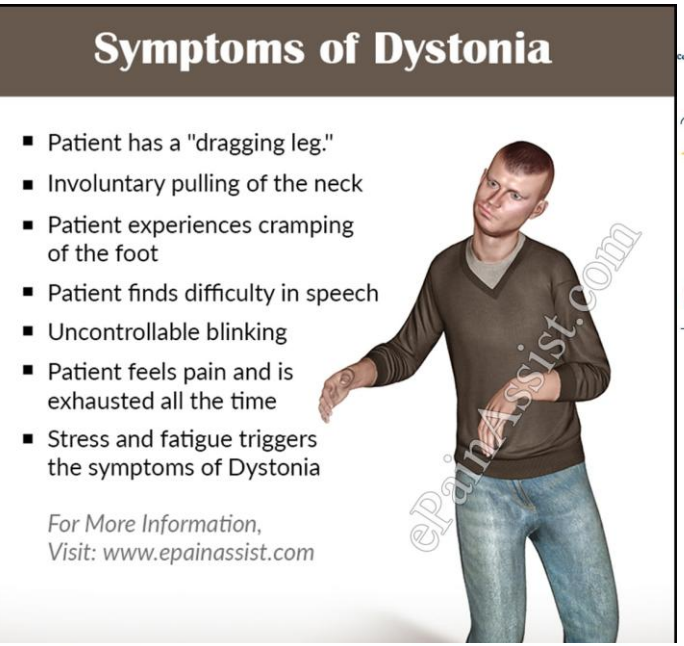
בקרה מוטורית

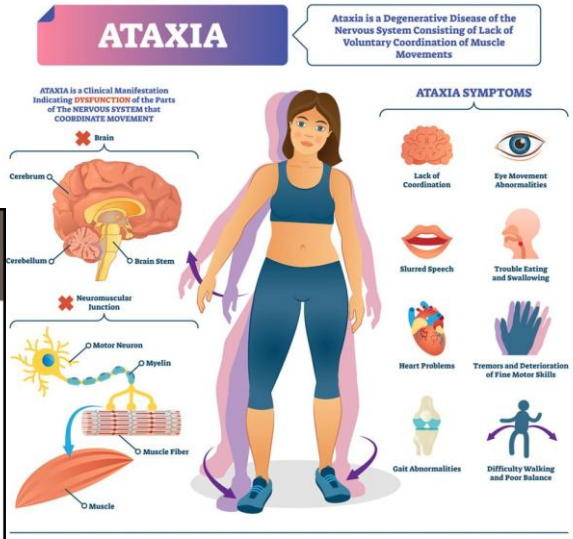


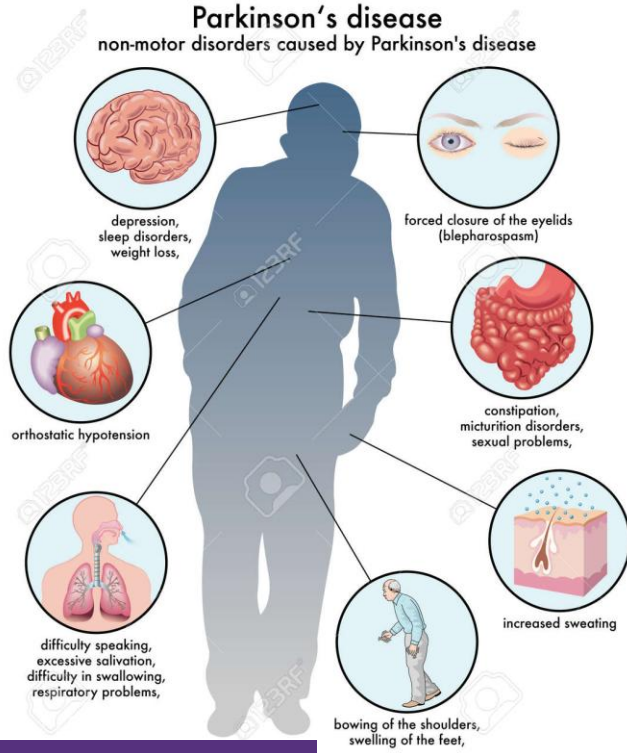




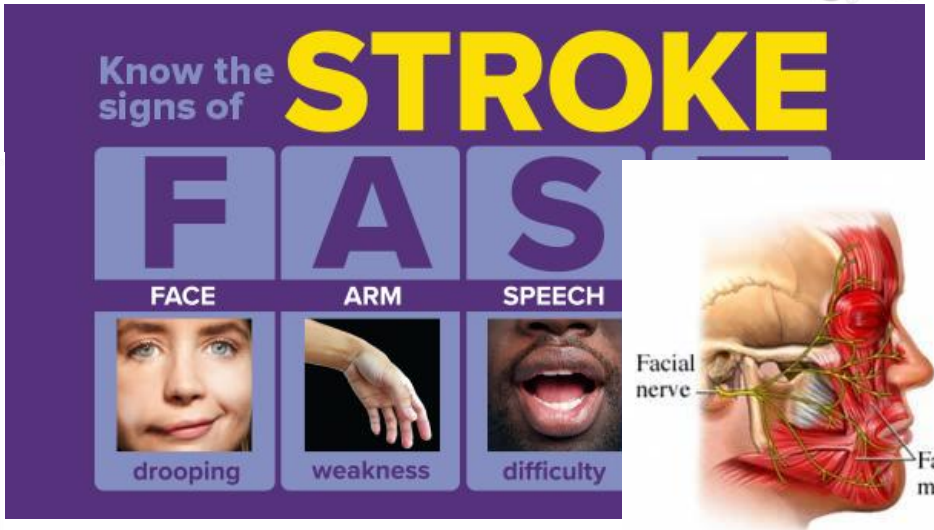
Pathologies of the motor system

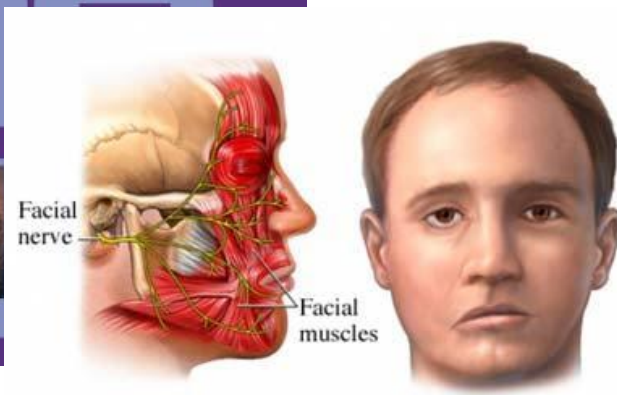












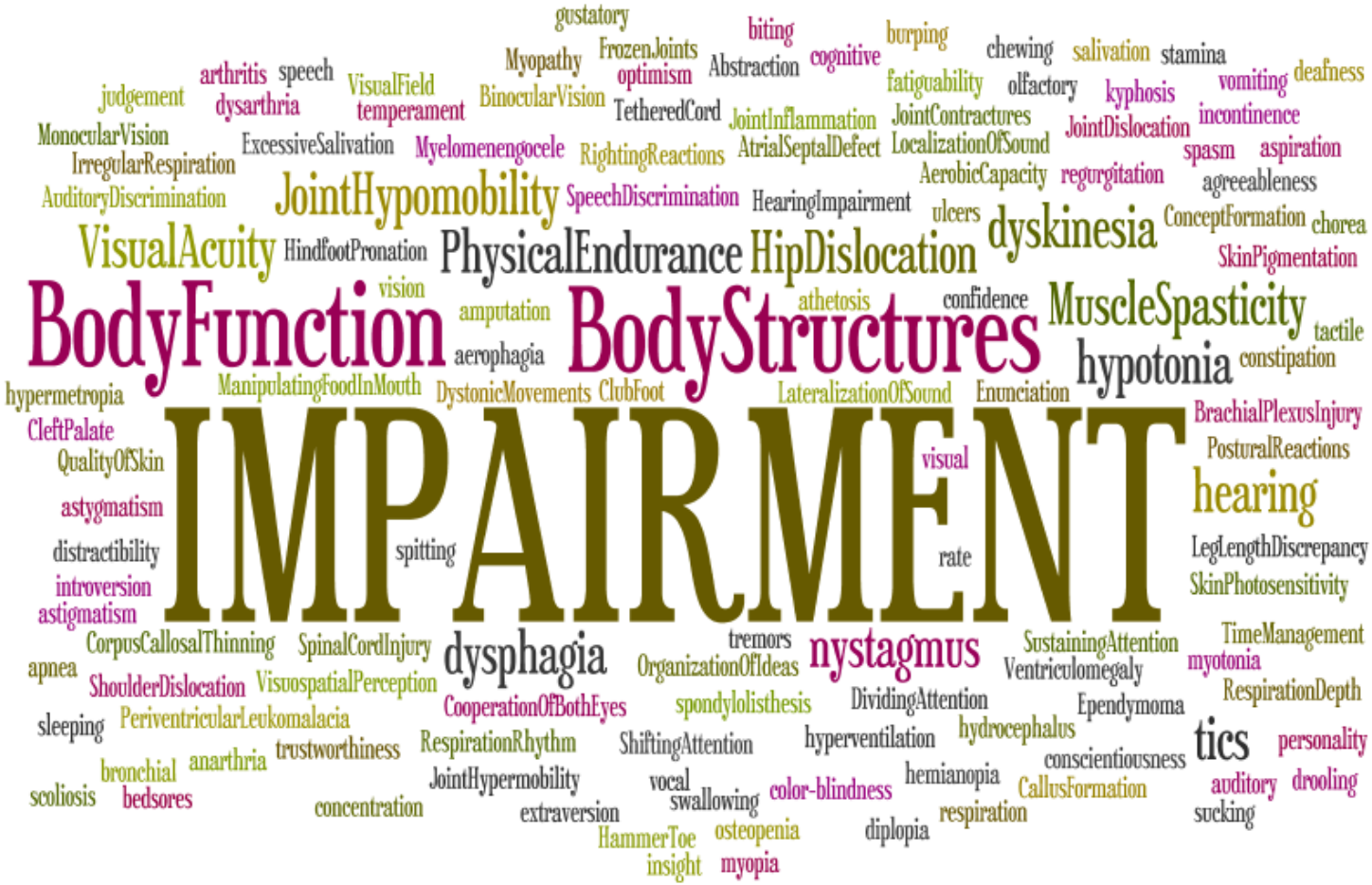
The Negev Lab



The Future Rehabilitation Hospital



What is impairment?



מטרות הקורס

- הכרת המערכת המוטורית
- דיון במקום של פתולוגיות במחקר במדעי המוח
- הכרת פתולוגיות מרכזיות של המערכת המוטורית
- הכרת פתולוגיות מרכזיות אשר קשורות בזקנה
- יכולת קריאת מאמרים ואינטגרציה
- חשיבה ביקורתית
- מיומנויות הצגה

חובות הקורס

- נוכחות והשתתפות.
- הצגת אחד הנושאים + כתיבת תקציר להצעת מחקר בנושא שהוצג.
- קריאה וכתיבת תגובה קצרה על מאמרים שיוצגו עד יום שלישי שלפני השיעור.

ביקור בעדי נגב?

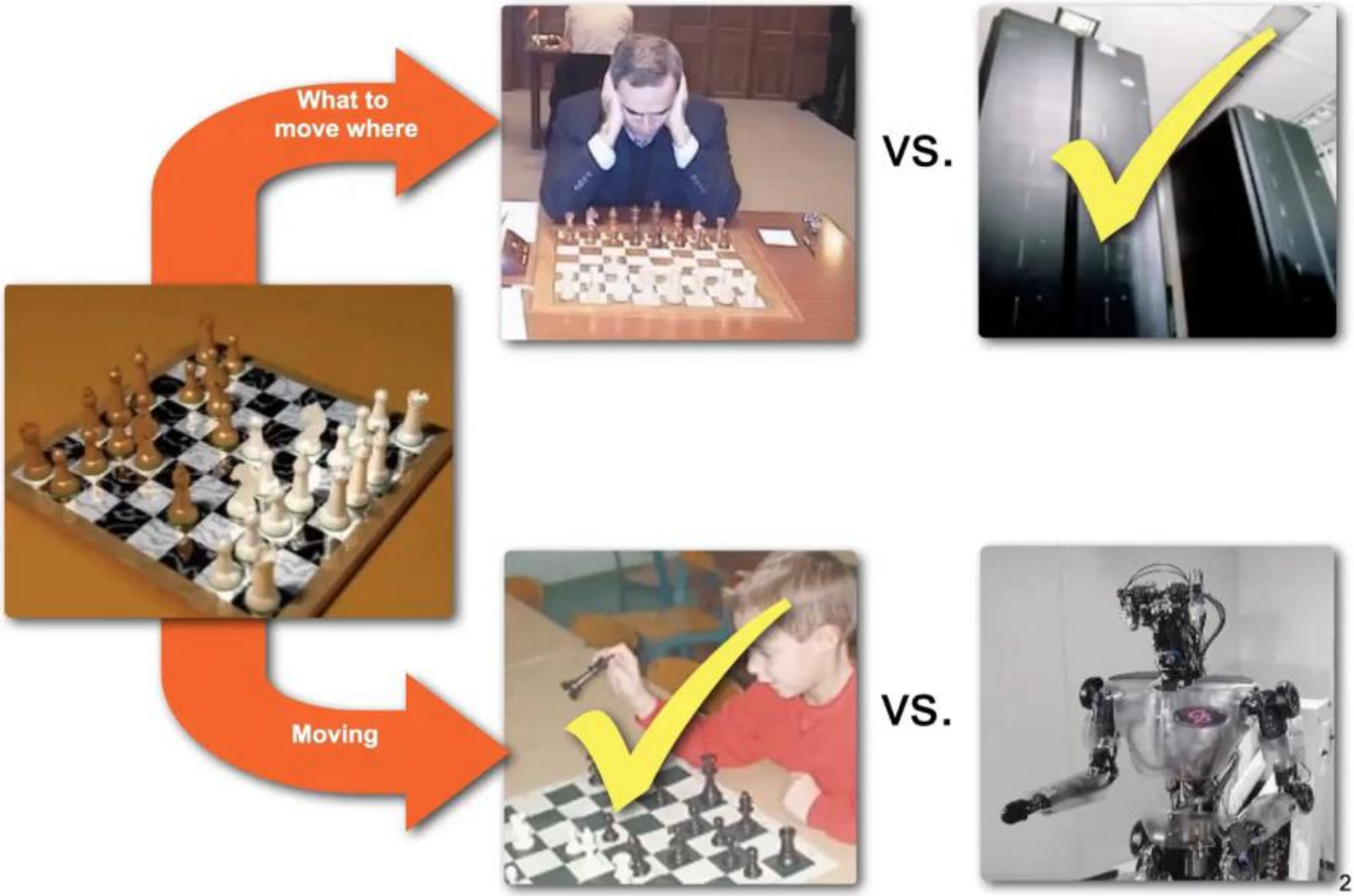
Theme	Week	
Introduction – Motor control	21.10	1
Introduction – Motor system	28.10	2
Stroke phenotype	4.11	3
Stroke brain	11.11	4
Ataxia	18.11	5
Parkinson's and Dystonia	25.11	6
The aging brain 1	2.12	7
The aging brain 2	9.12	8
Pathologies of aging : FTD	16.12	9
Pathologies of aging : AD	23.12	10
Pathologies of aging : Cognitive decline	30.12	11
Pain	6.1	12
Adi Negev visit?	13.1	13

The goal of the brain is to move



Sea squirt

Complexity of human movement control



Taken from Wolpert's talk

Motor control covers a variety of behaviors













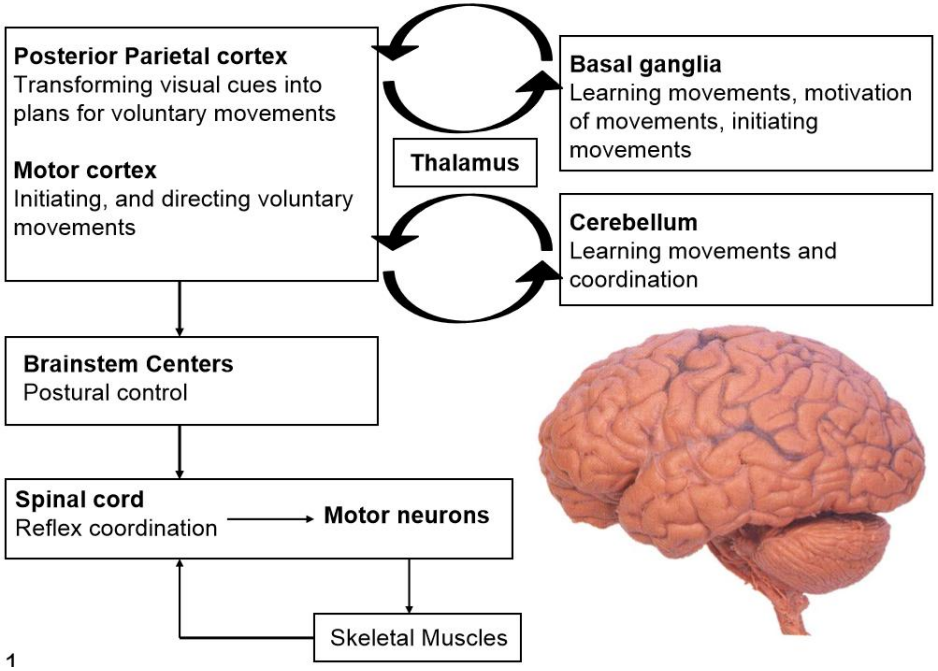


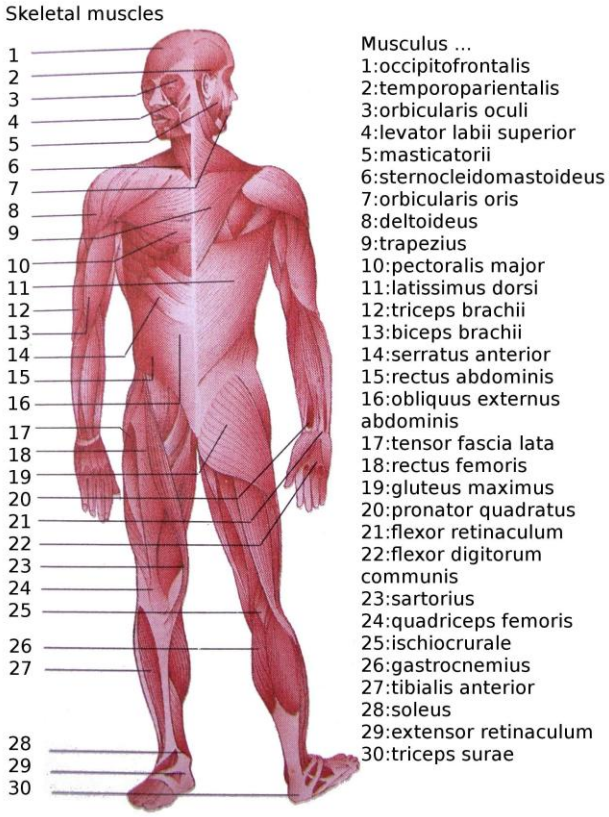






There is a very complex underlying machinery





So how should we study the motor system?



I. Levels of Control

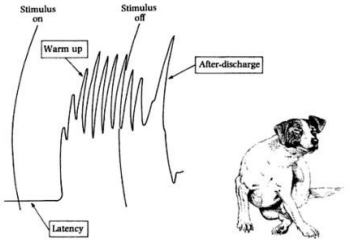
- Voluntary movements

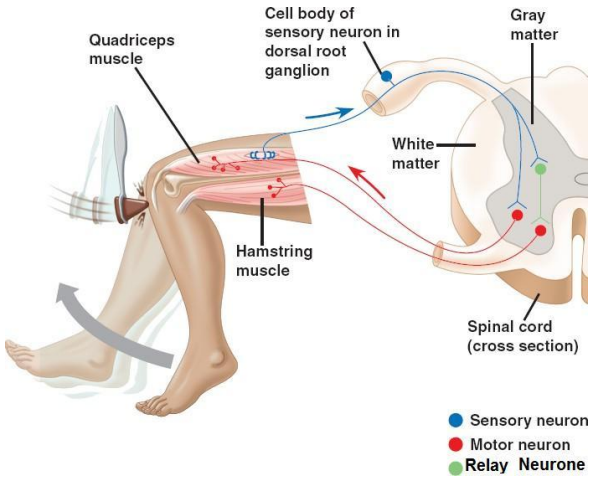


- Rhythmic movements



- Reflexes
  - Stretch
  - Scratch





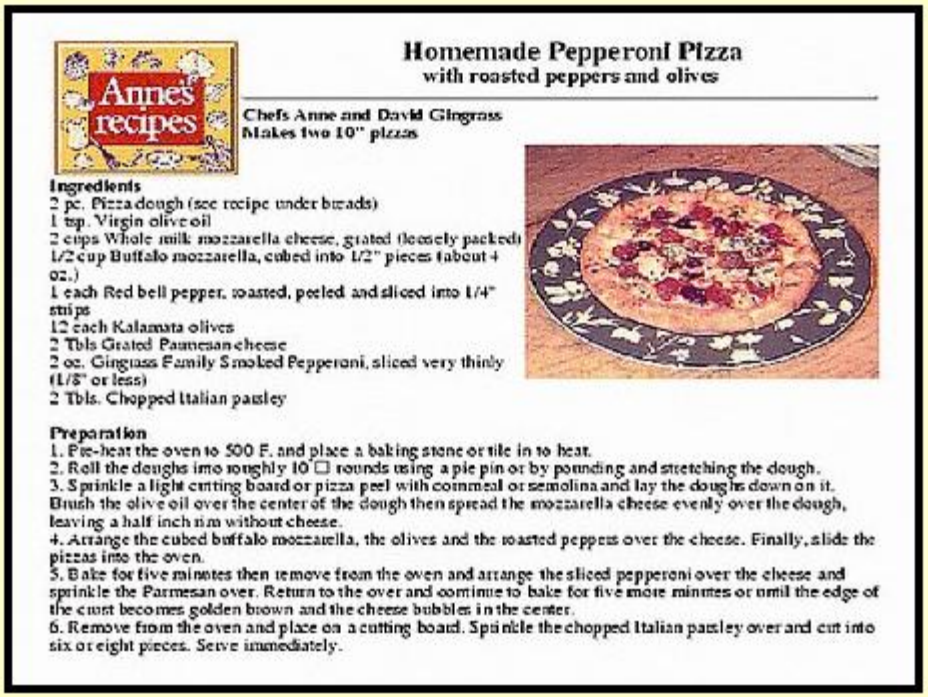
II. Marr's Tri-Level Hypothesis

- Computational level –
  - what does the system do?
  - what is the purpose of an action?
- Algorithmic level –
  - how does the system do it?
- Implementation level –
  - how does the system realizes this computation?

MEANS

METHOD

GOAL



Anne's recipes

Homemade Pepperoni Pizza
with roasted peppers and olives

Chefs Anne and David Gingrass
Makes two 10" pizzas

Ingredients
2 pc. Pizza dough (see recipe under breads)
1 tsp. Virgin olive oil
2 cups Whole milk mozzarella cheese, grated (loosely packed)
1/2 cup Buffalo mozzarella, cubed into 1/2" pieces (about 4 oz.)
1 each Red bell pepper, roasted, peeled and sliced into 1/4" strips
12 each Kalamata olives
2 Tbls Grated Parmesan cheese
2 oz. Gingrass Family Smoked Pepperoni, sliced very thinly (1/8" or less)
2 Tbls. Chopped Italian parsley

Preparation
1. Pre-heat the oven to 500 F. and place a baking stone or tile in to heat.
2. Roll the doughs into roughly 10" rounds using a pie pin or by pounding and stretching the dough.
3. Sprinkle a light cutting board or pizza peel with cornmeal or semolina and lay the doughs down on it. Brush the olive oil over the center of the dough then spread the mozzarella cheese evenly over the dough, leaving a half inch rim without cheese.
4. Arrange the cubed buffalo mozzarella, the olives and the roasted peppers over the cheese. Finally, slide the pizzas into the oven.
5. Bake for five minutes then remove from the oven and arrange the sliced pepperoni over the cheese and sprinkle the Parmesan over. Return to the over and continue to bake for five more minutes or until the edge of the crust becomes golden brown and the cheese bubbles in the center.
6. Remove from the oven and place on a cutting board. Sprinkle the chopped Italian parsley over and cut into six or eight pieces. Serve immediately.

GOAL Origin of species
(Darwin, 1844)
Evolution theory, survival of the fittest



METHOD Experiments in plant hybridization
(Mendel, 1865)
Heredity theory, classical genetics



MEANS Molecular structure of DNA
(Watson & Crick, 1953)
Molecular biology, modern genetics



Taken from Peter Arnold van der Helm

Can we study processes by examination of the processor?

Neuron
Perspective

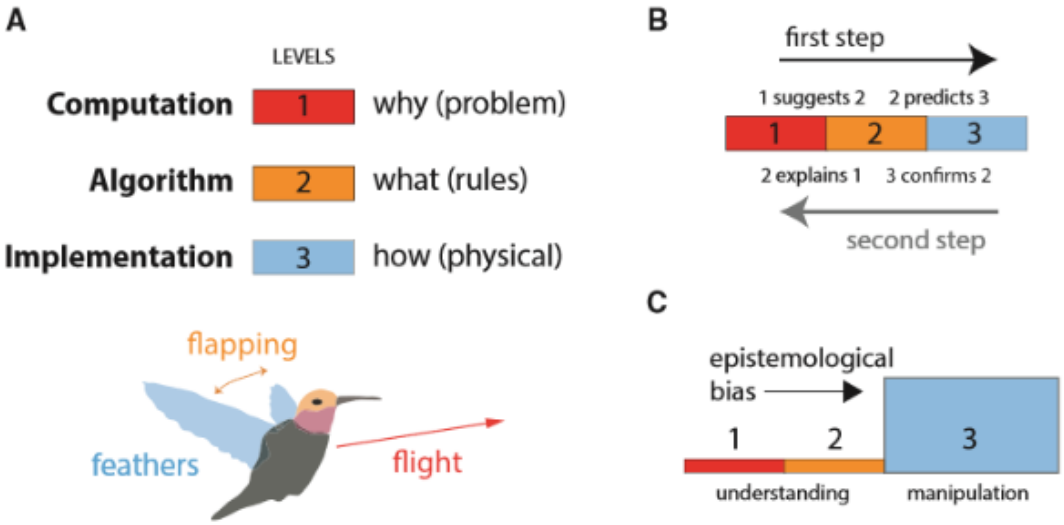


Figure 2. Marr's Three Levels of Analysis
(A) A bird attempts to fly (goal) by flapping its wings (algorithmic realization) whose aerodynamics depend on the features of its feathers (physical implementation). Feathers "have something to do" with flight and flapping, but what level of understanding do we achieve if we dissect the properties of the feathers alone? Bats fly but don't have feathers, and birds can fly without flapping.
(B) The relationship between the three levels is not arbitrary; step 1 comes before step 2: the algorithmic level of understanding is essential to interpret its mechanistic implementation. Step 2: implementation level work feeds back to inform the algorithmic level.
(C) An epistemological bias toward manipulation-based view of understanding induced by technology (black filled arrow).

Krakauer et al., 2016

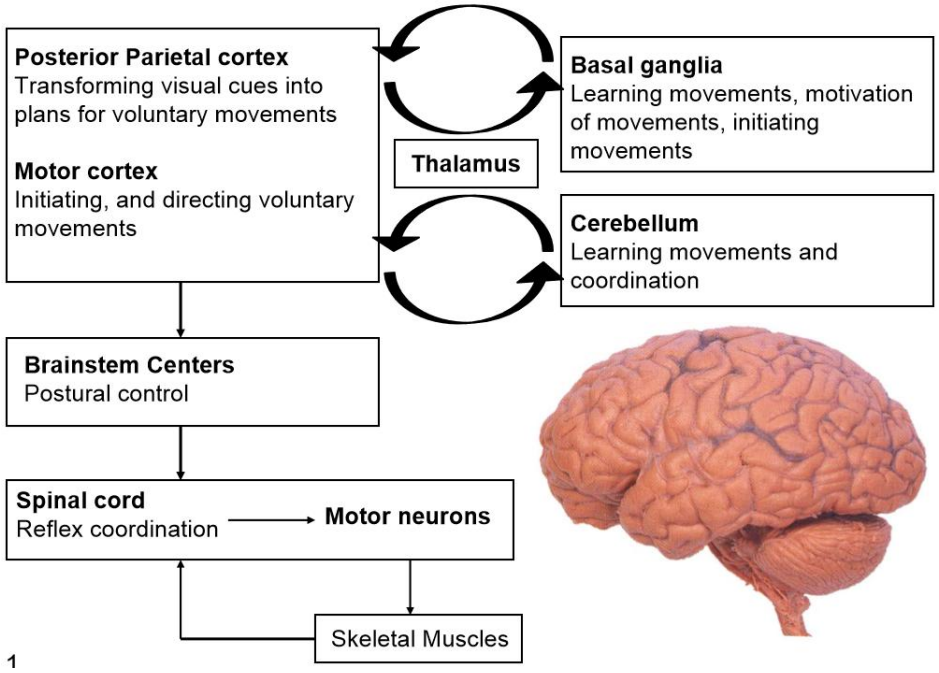
Trying to understand perception by understanding neurons is like trying to understand bird's flight by studying only feathers. It just cannot be done (Marr 1982)

The substitution bias

When faced with a difficult question, we often answer an easier one instead, usually without noticing the substitution (Kahneman 2011).

III. How should we study the motor system?

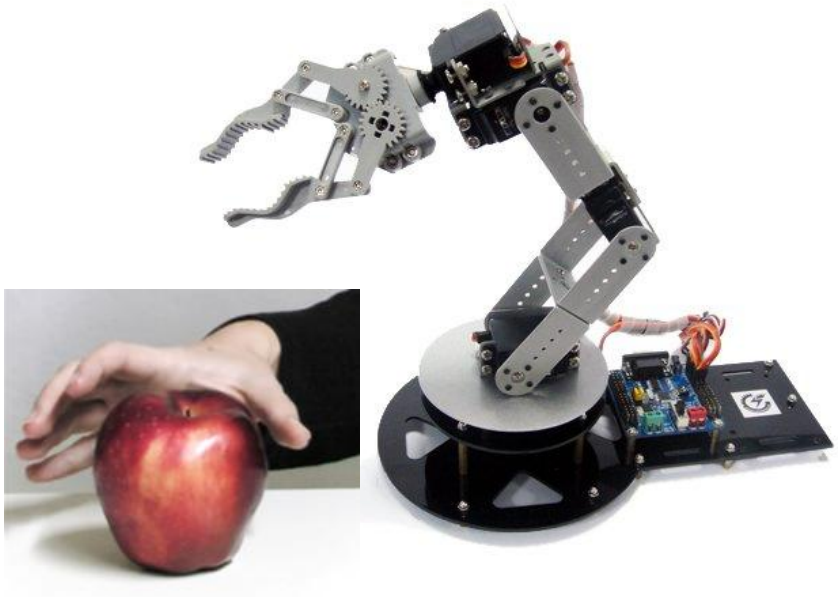
Study of the motor circuitry



1

How do we do that?

Computational motor control



Why do we reach in the way that we do?

Computational motor control

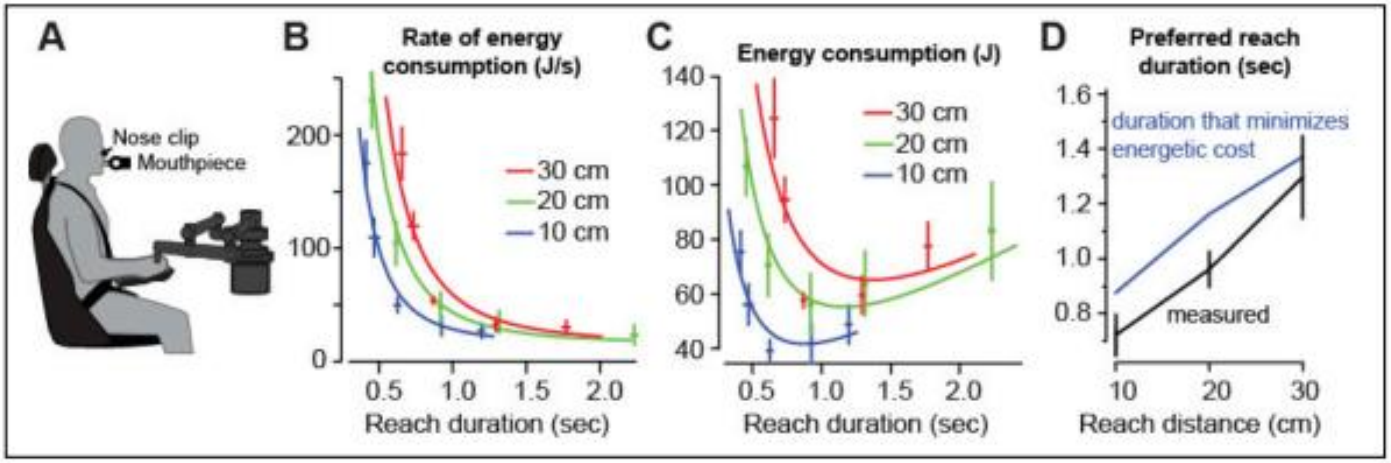
- The goal is to produce laws that describe regularities in movements
- Natural movements
- Movements under perturbations
- Equations that are independent from the hardware

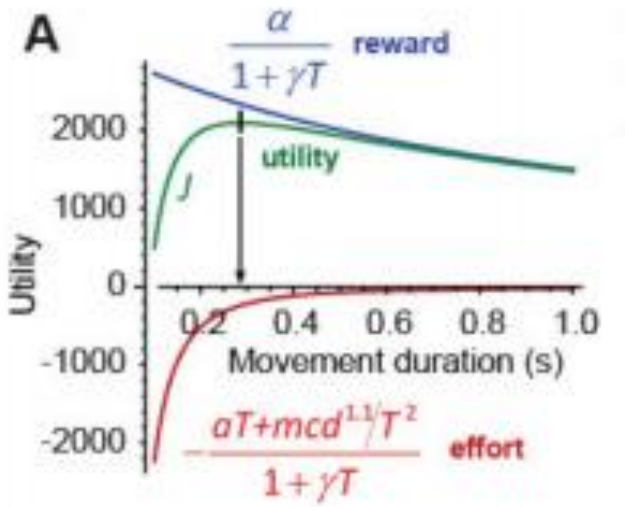
II. Marr's Tri-Level Hypothesis



- Computational level –
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  - how does the system realizes this computation?

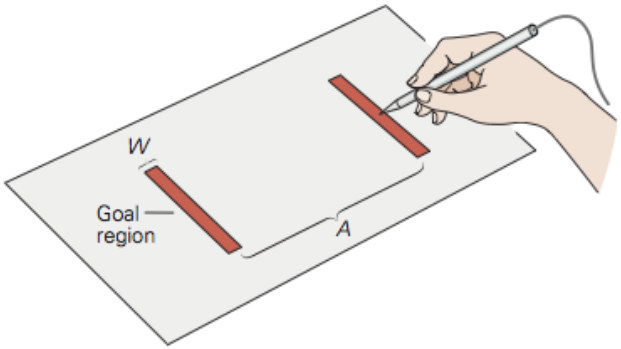
Computational level: minimize energy consumption?

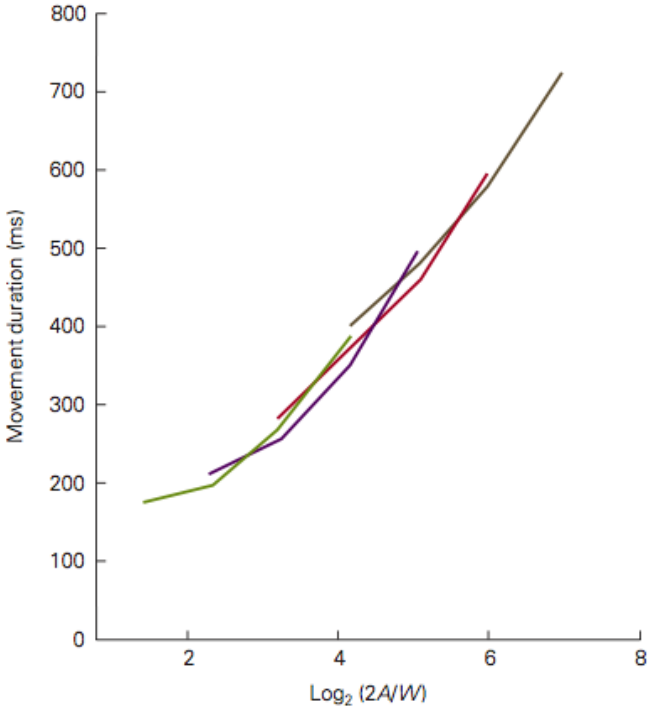




Reza Shadmehr and Alaa Ahmed

Fitt's law (1954)

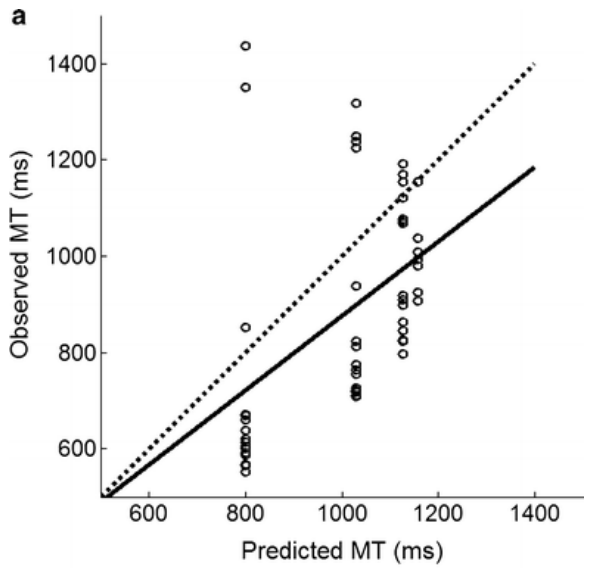


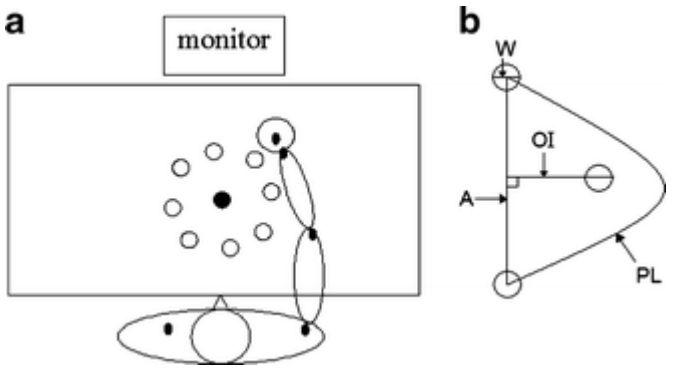


$$T = a + b \log_2 \left(1 + \frac{D}{W}\right)$$

(Jeannerod 1988)

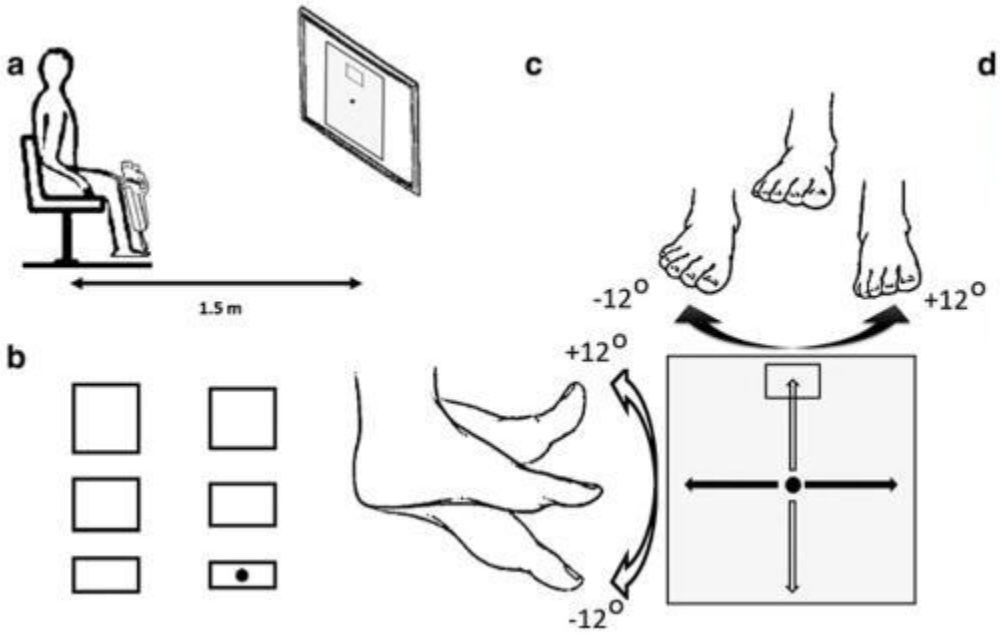
Fitt's law in obstacle avoidance

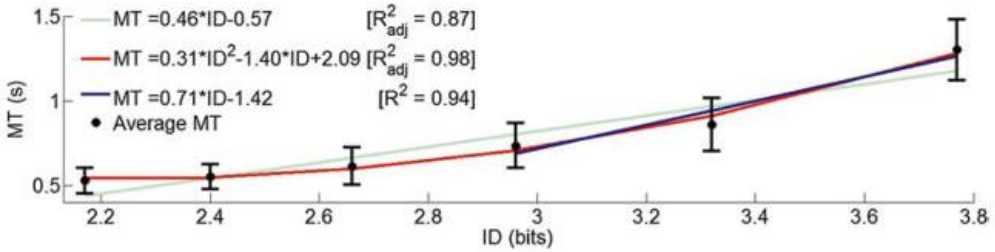




Jax et al., 2007

Fitt's law in ankle movements





Michmizos and Krebs 2014

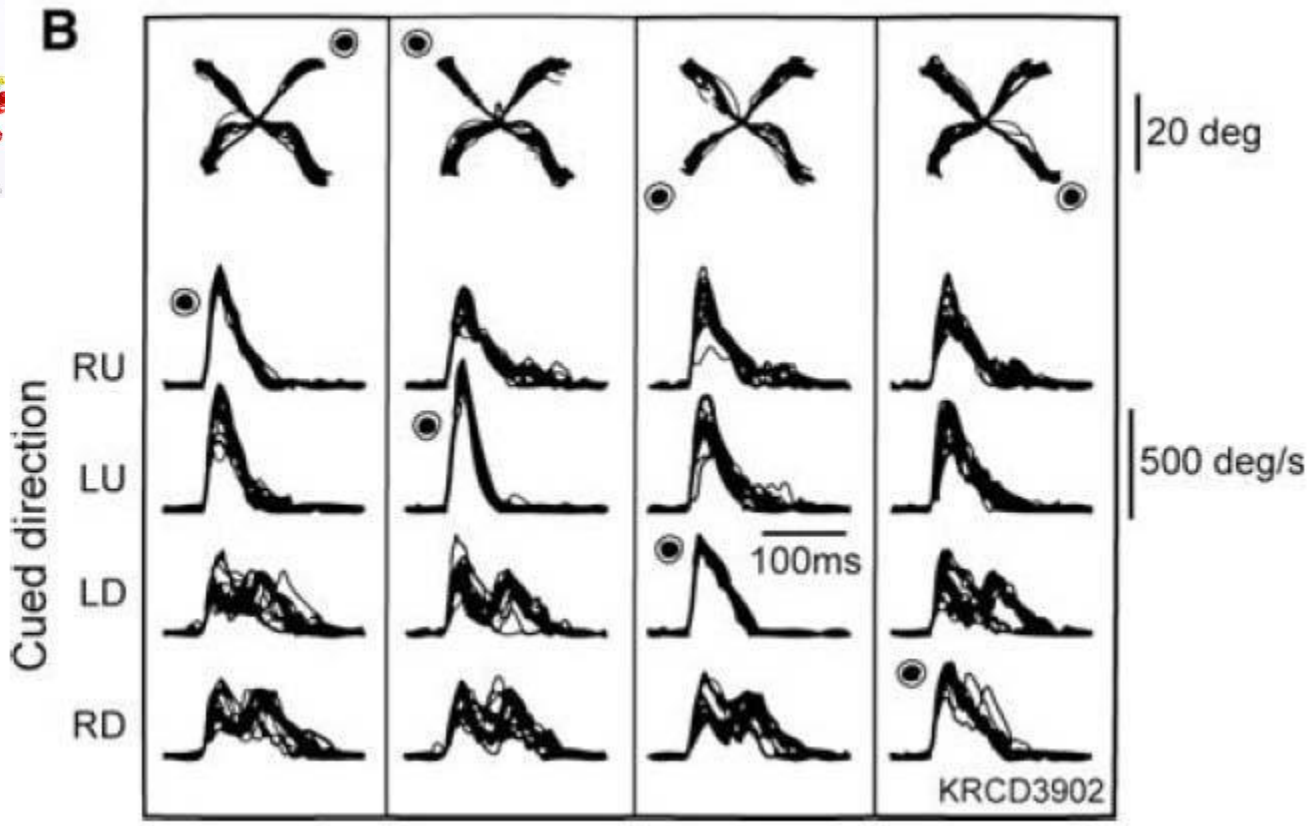
Movement planning is affected by accuracy demands and by the required forces

Linking variability and planning

- Planning includes optimization of the tradeoff between accuracy and size of motor command (why do we want the size of the motor command to be higher?)
- Larger motor command → higher neural activity → noisier movements → lower accuracy
- Solution could not be fixed since costs and rewards change

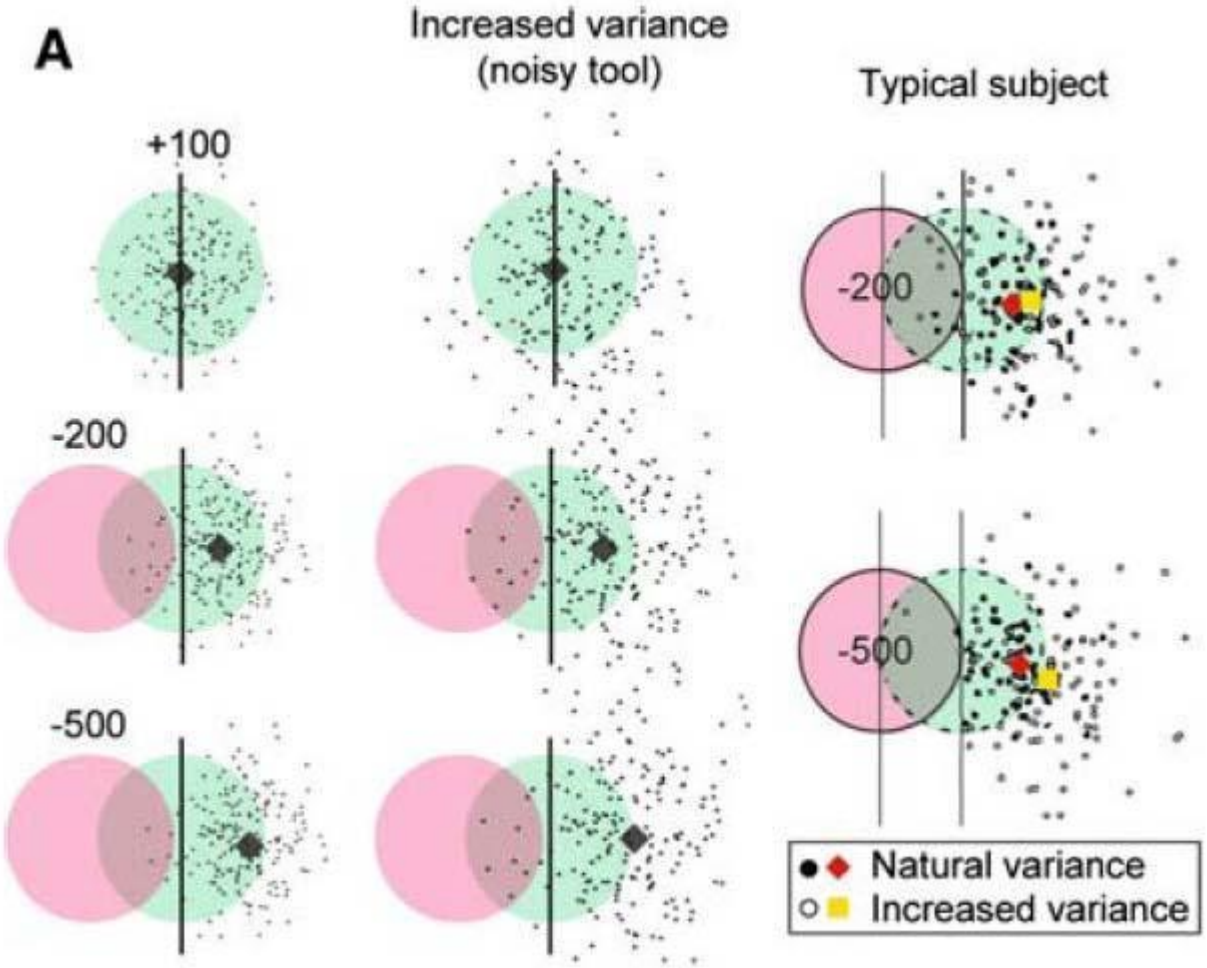


Reward affects kinematics



Takikawa et al., 2002

Variability is taken into account when planning a movement



Trommershauser et al., 2005

Interim summary - optimization

Actions are selected based on optimization criteria:

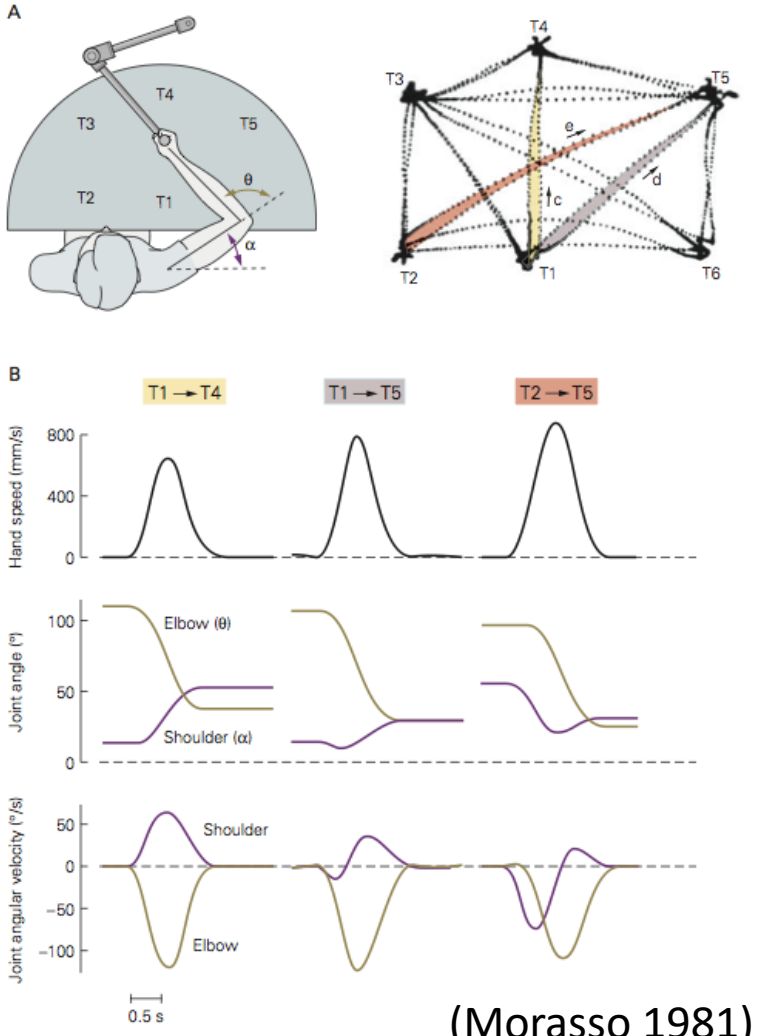
time and energy

accuracy and velocity (force)

reward and effort

Regularities in trajectories

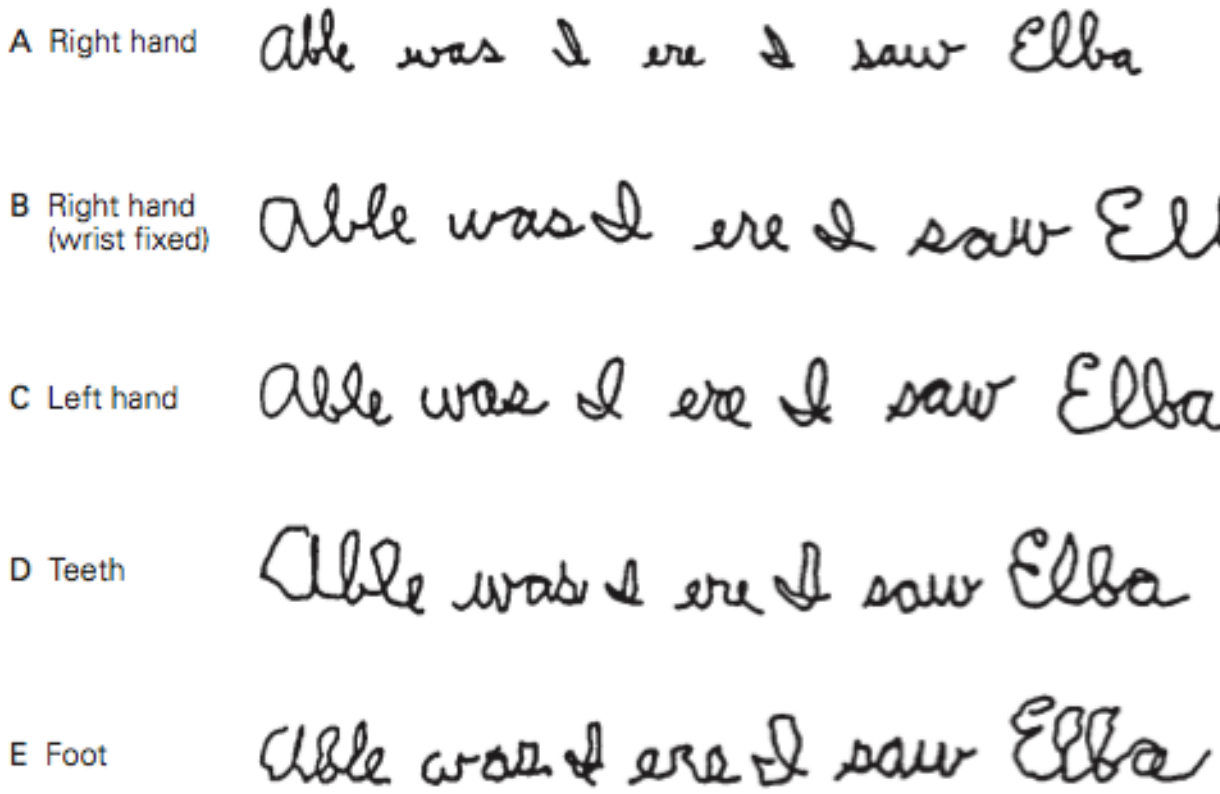
1. Movements are straight.
2. Smoothed, unimodal and symmetric velocity profile.
3. Hand trajectories are more invariant than joint trajectories
4. Why?



(Morasso 1981)

Movement are planned at the level of hand trajectory

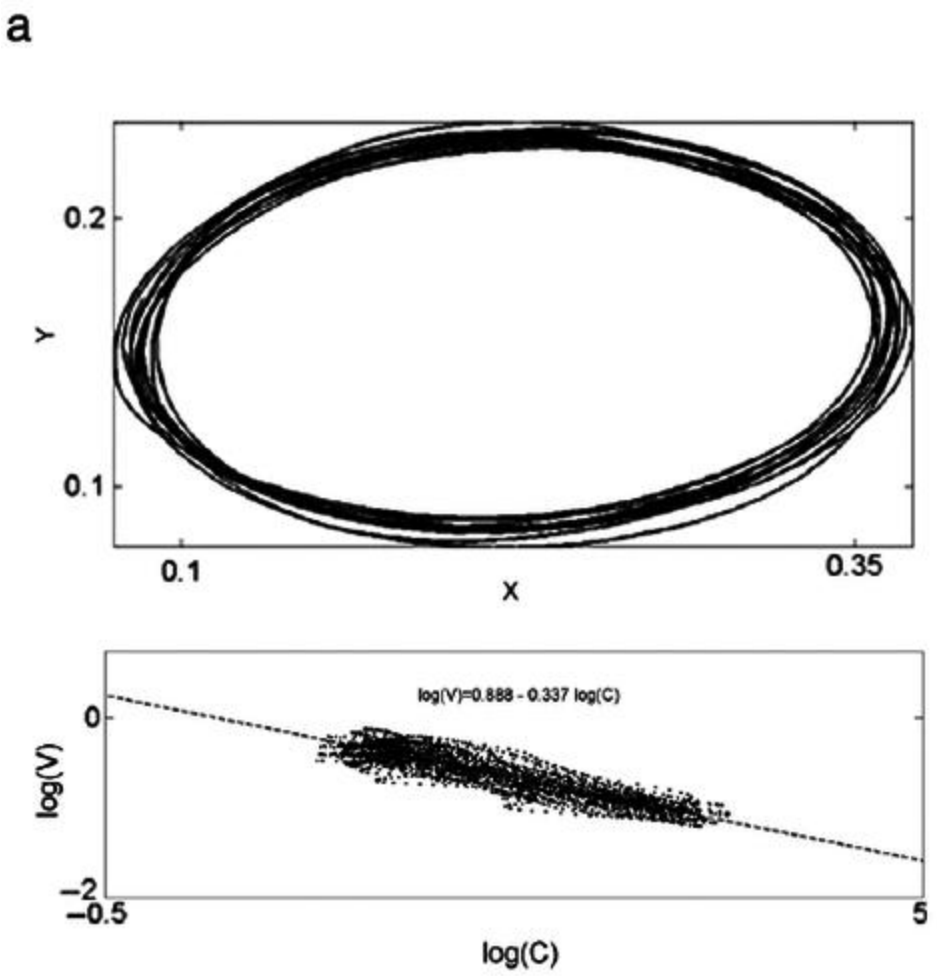
Implementation: Motor equivalence

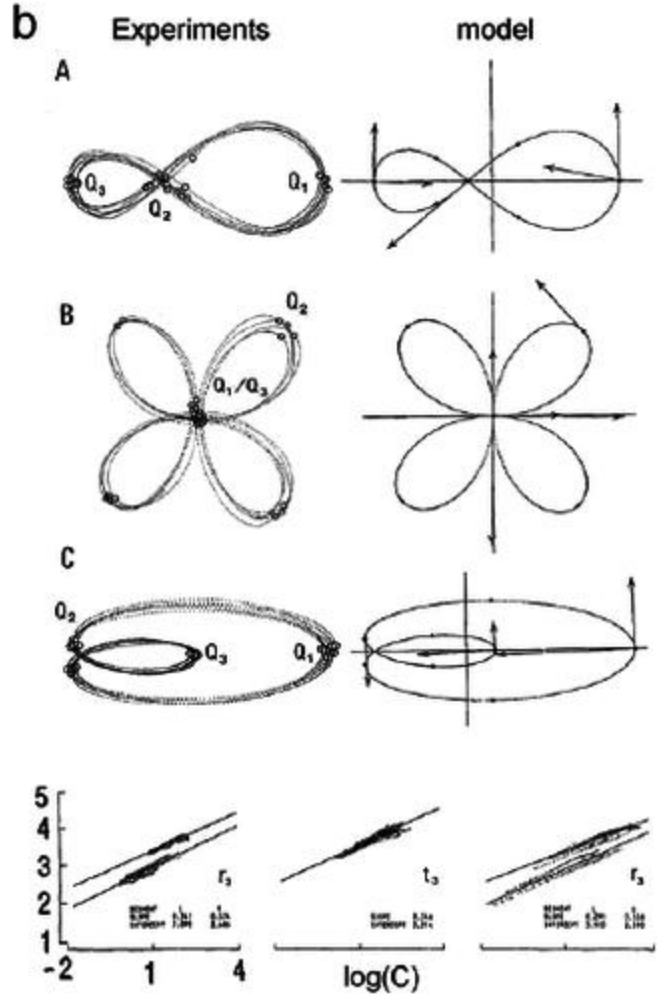


Abstract (effector independent) representation of movements

Raibert 1977

Velocity and Curvature are related





Viviani and Flash 1995

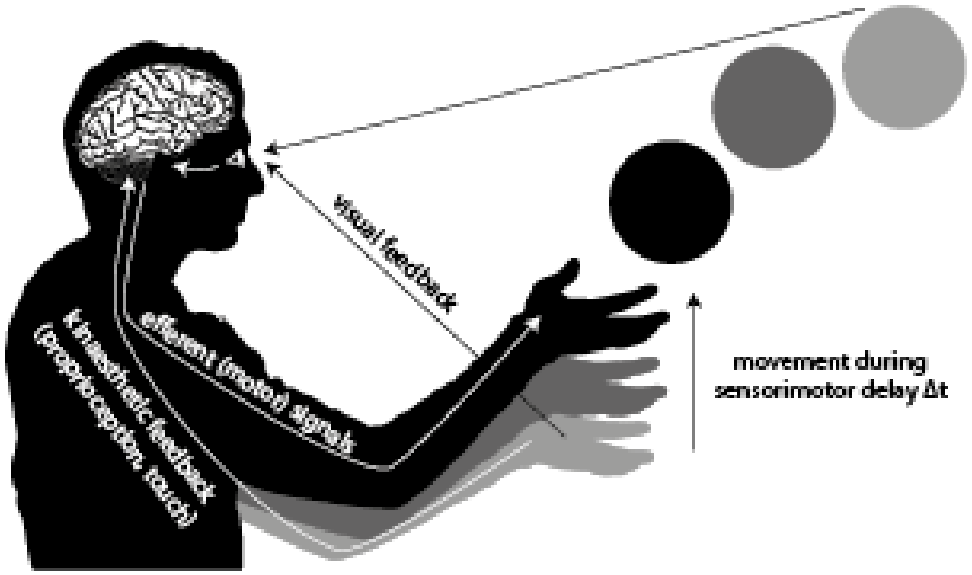
Interim summary - Implementation

Movements are straight in visual space

Representations are shared across effectors

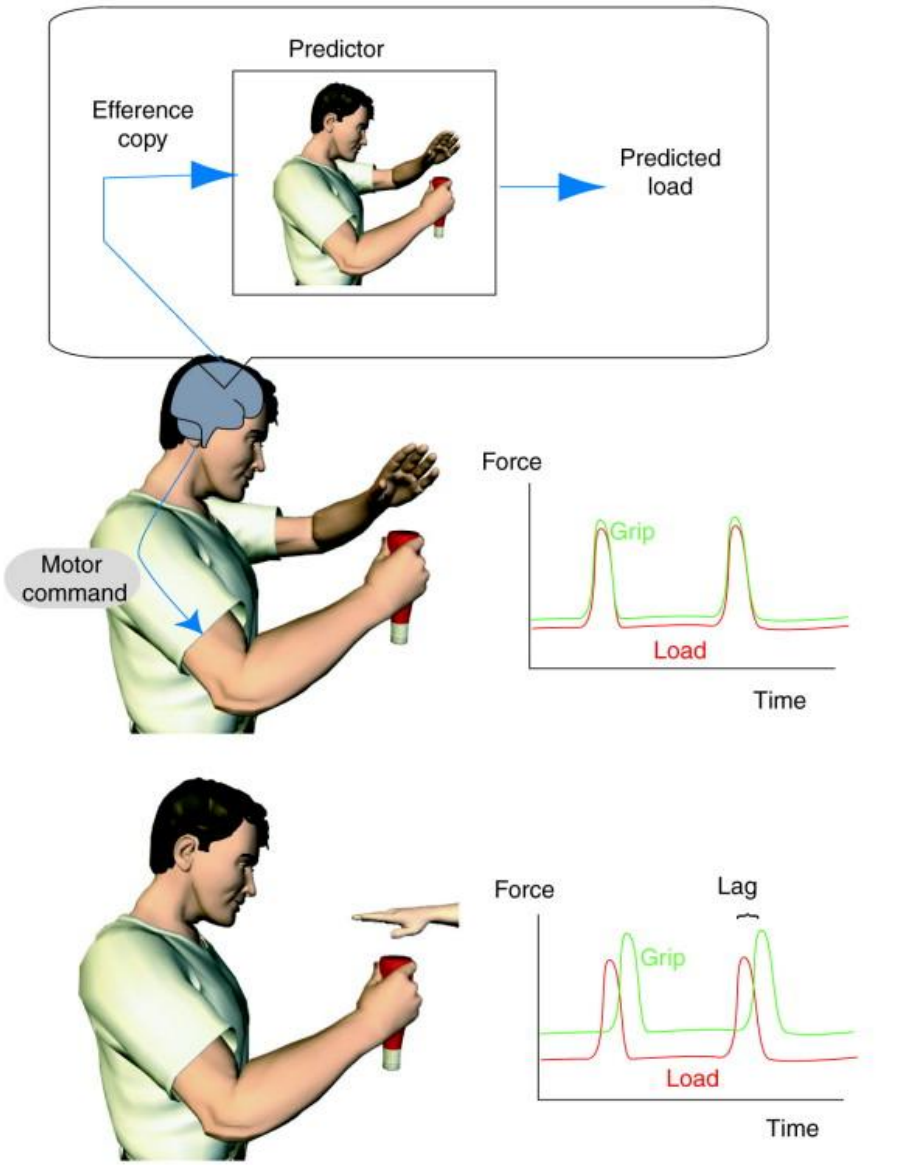
Noise is controlled across multiple movement parameters

Sensorimotor delays



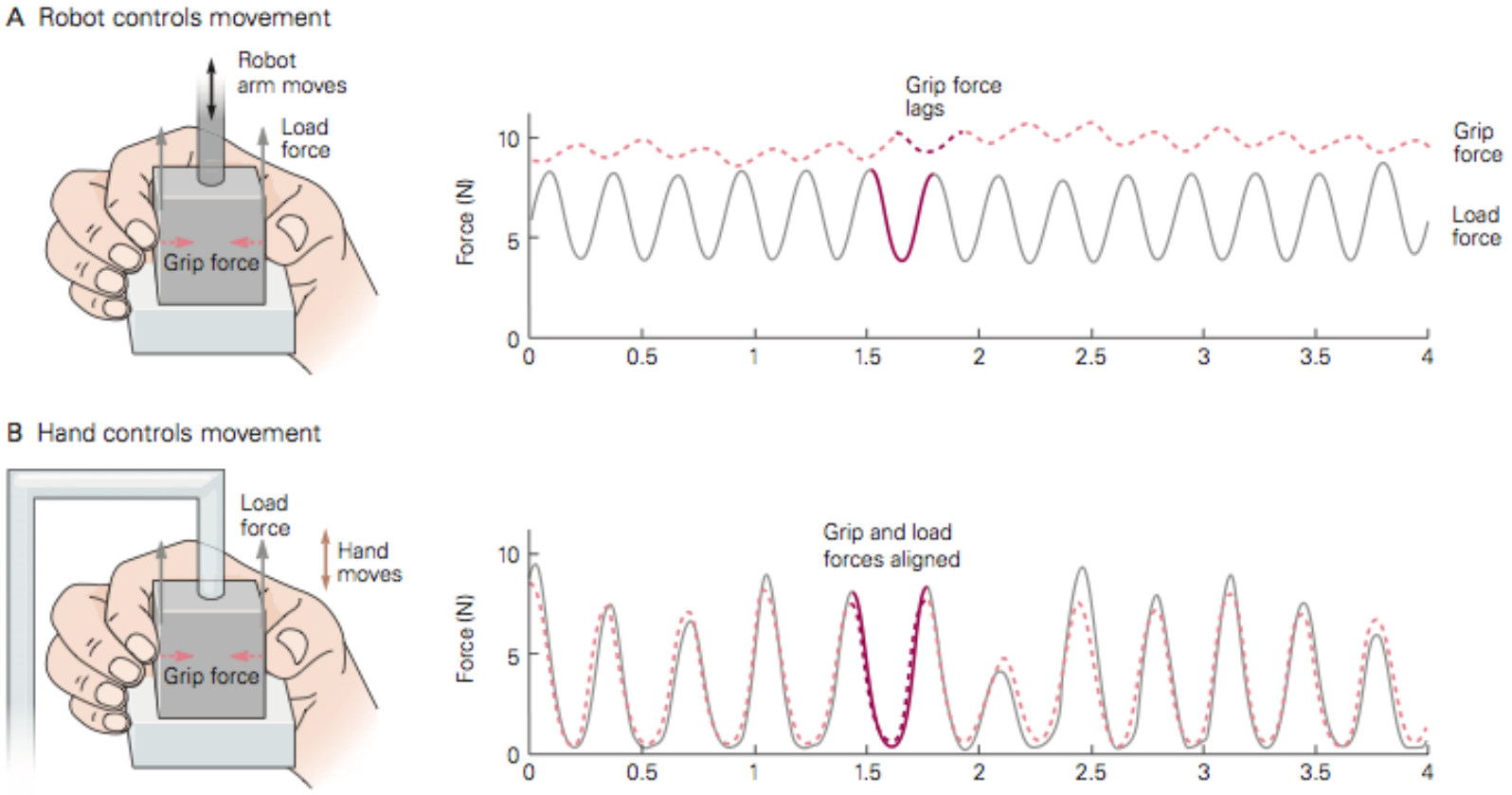
Kenneth Craik (1943)

"If the organism carries a 'small-scale model' of external reality and of its own possible actions within its head, it is able to try out various alternatives, conclude which is the best of them, react to future situations before they arise, utilize the knowledge of past events in dealing with the present and future, and in every way to react in a much fuller, safer, and more competent manner to the emergencies which face it."



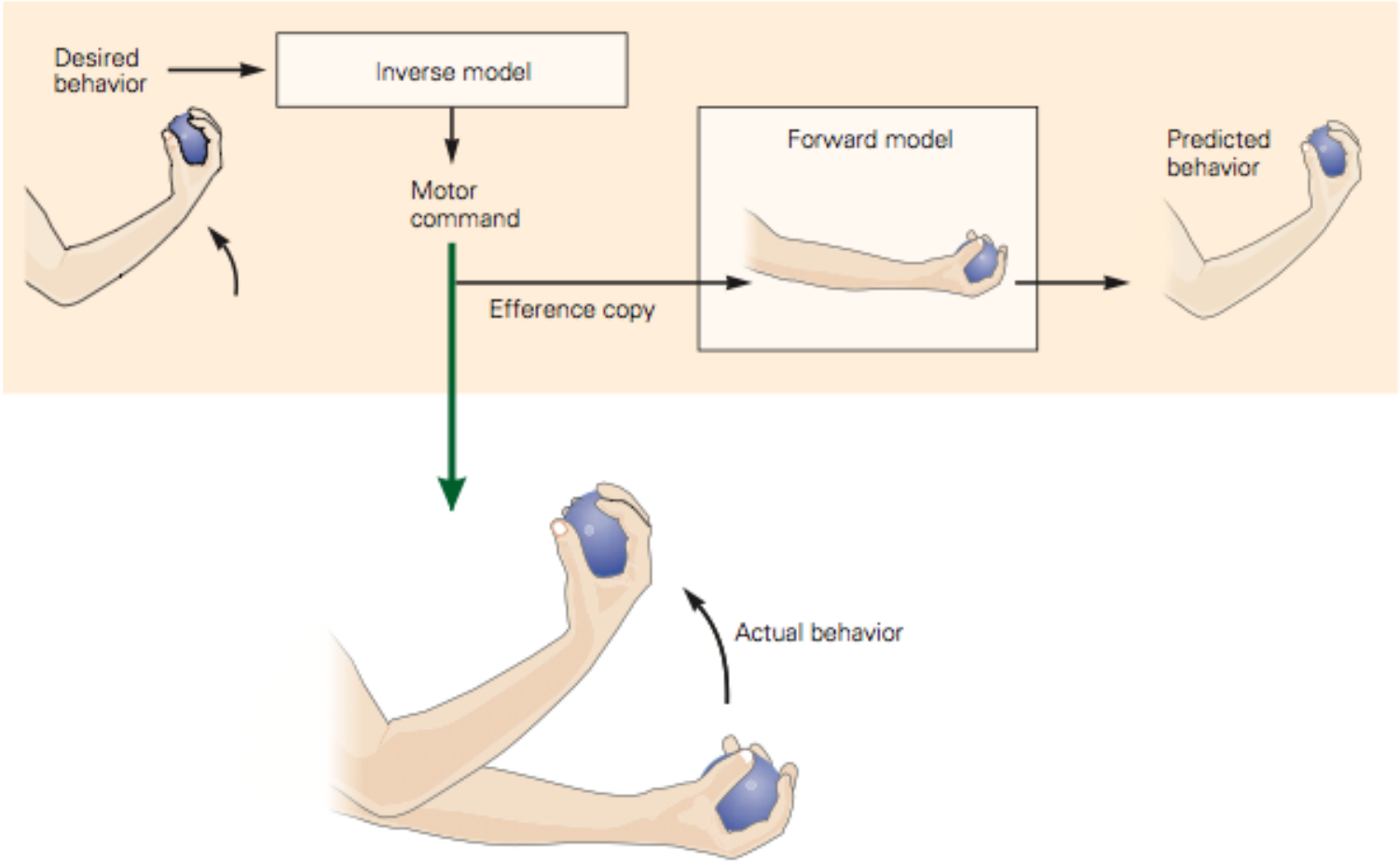
Wolpert and Flanagan 2001

Improved reaction for self generating movements



Blakemore et al., 1998

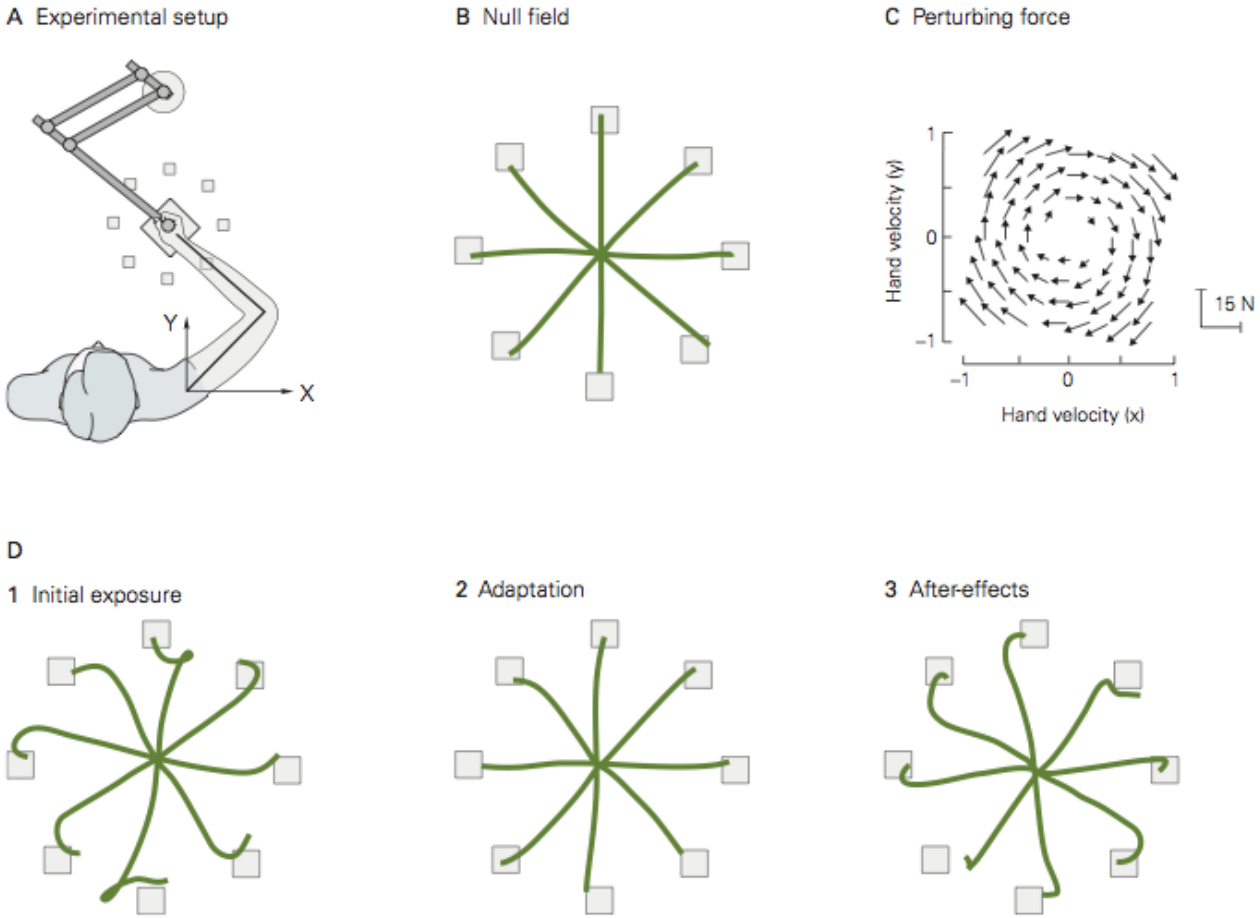
Internal models



Acting in a changing environment

Plan is determined by the dynamics of the body and tool, dynamics that change all the time and needs to be monitored

The relationship between body and command changes all the time



(Brashers-Krug, Shadmehr, and Bizzi 1996)

Key points

- Optimization
- Planning (Regularities)
- Delays
- Internal models
- Calibration

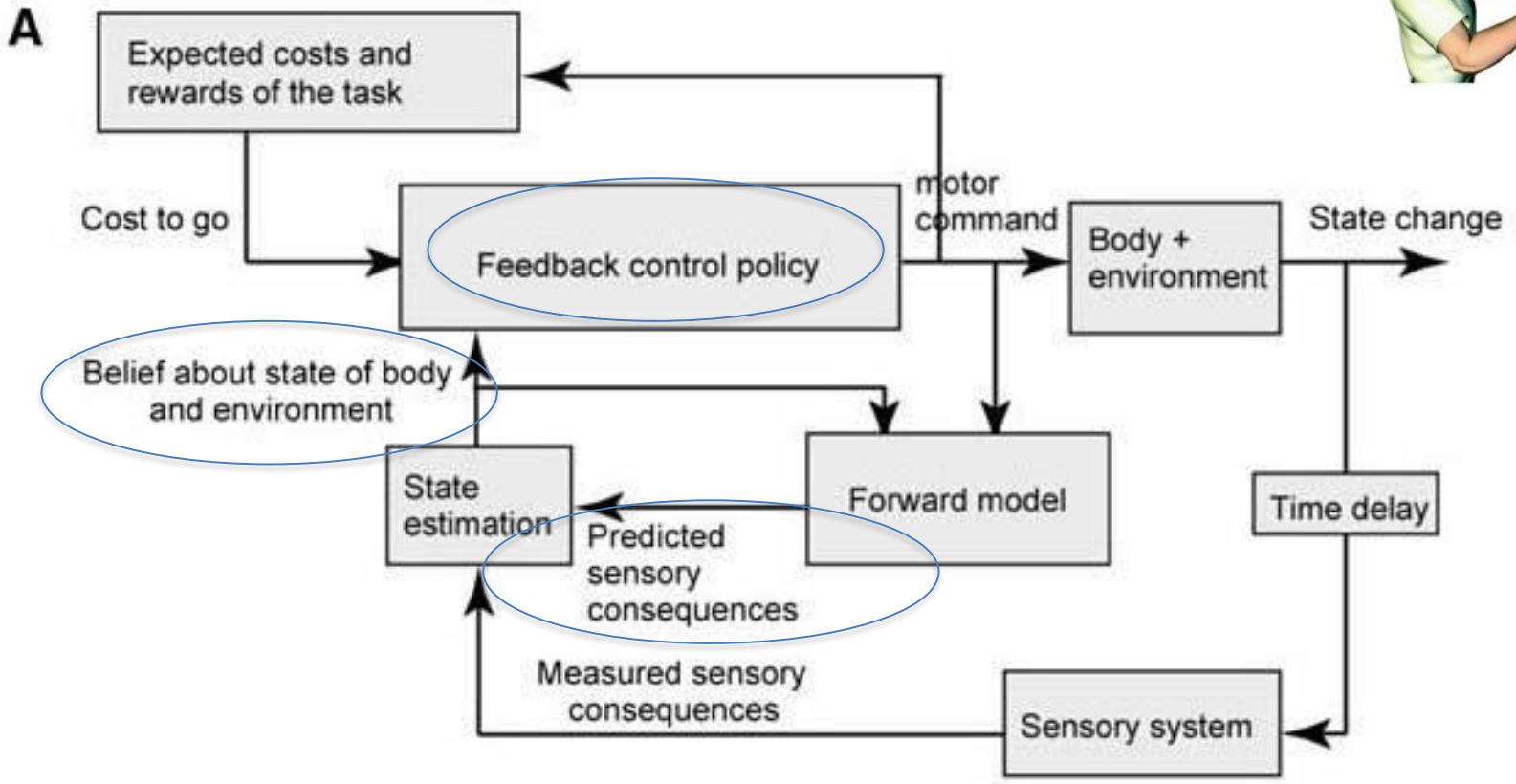
Unified theory for motor control

- Goal: optimization of costs and reward
- Implementation:
  - feedforward and feedback control
  - Internal models
  - Continues Calibration

Optimal feedback control

(Todorov and Jordan 2002)





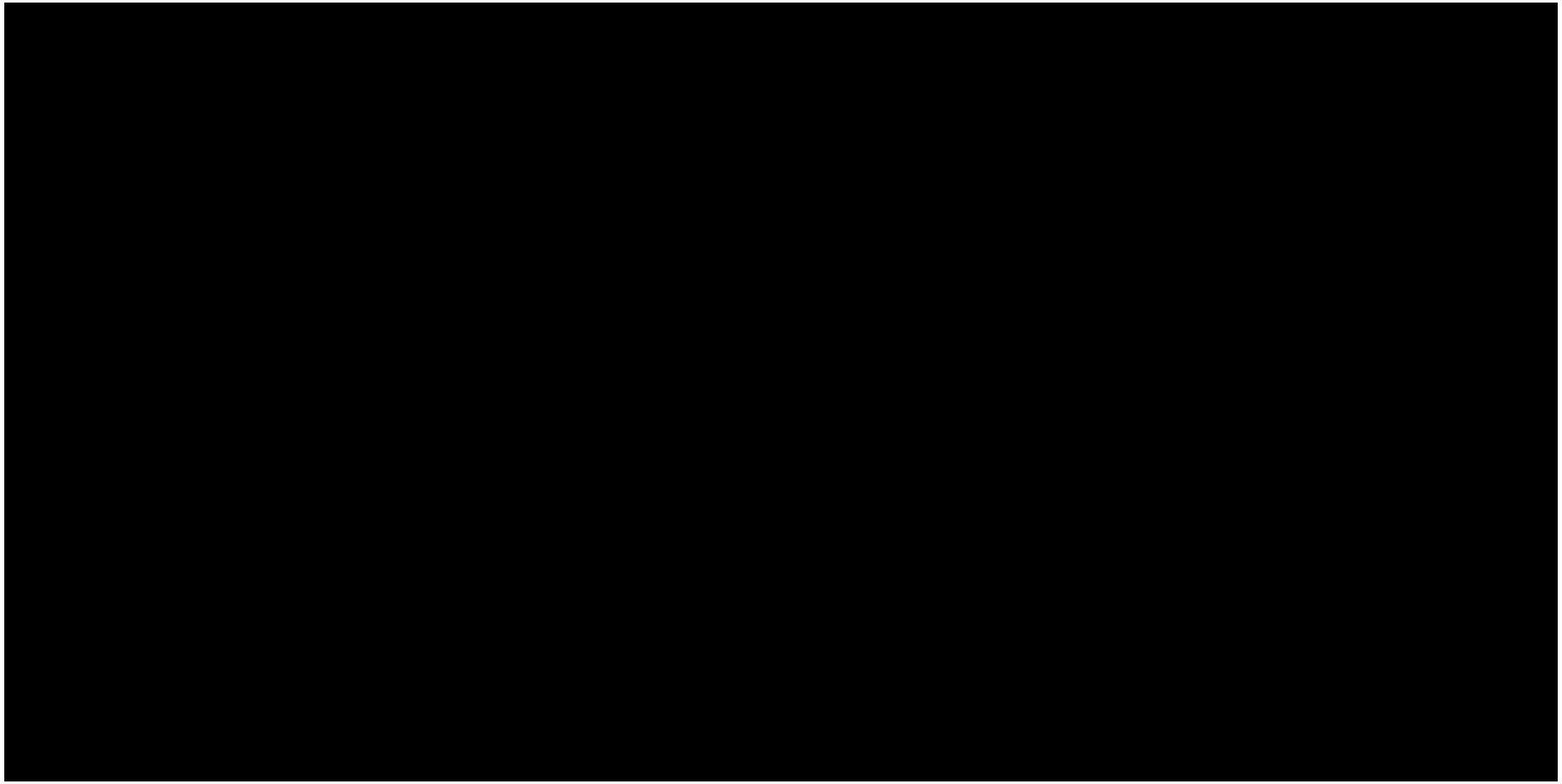
Krakauer and Shadmehr 2007

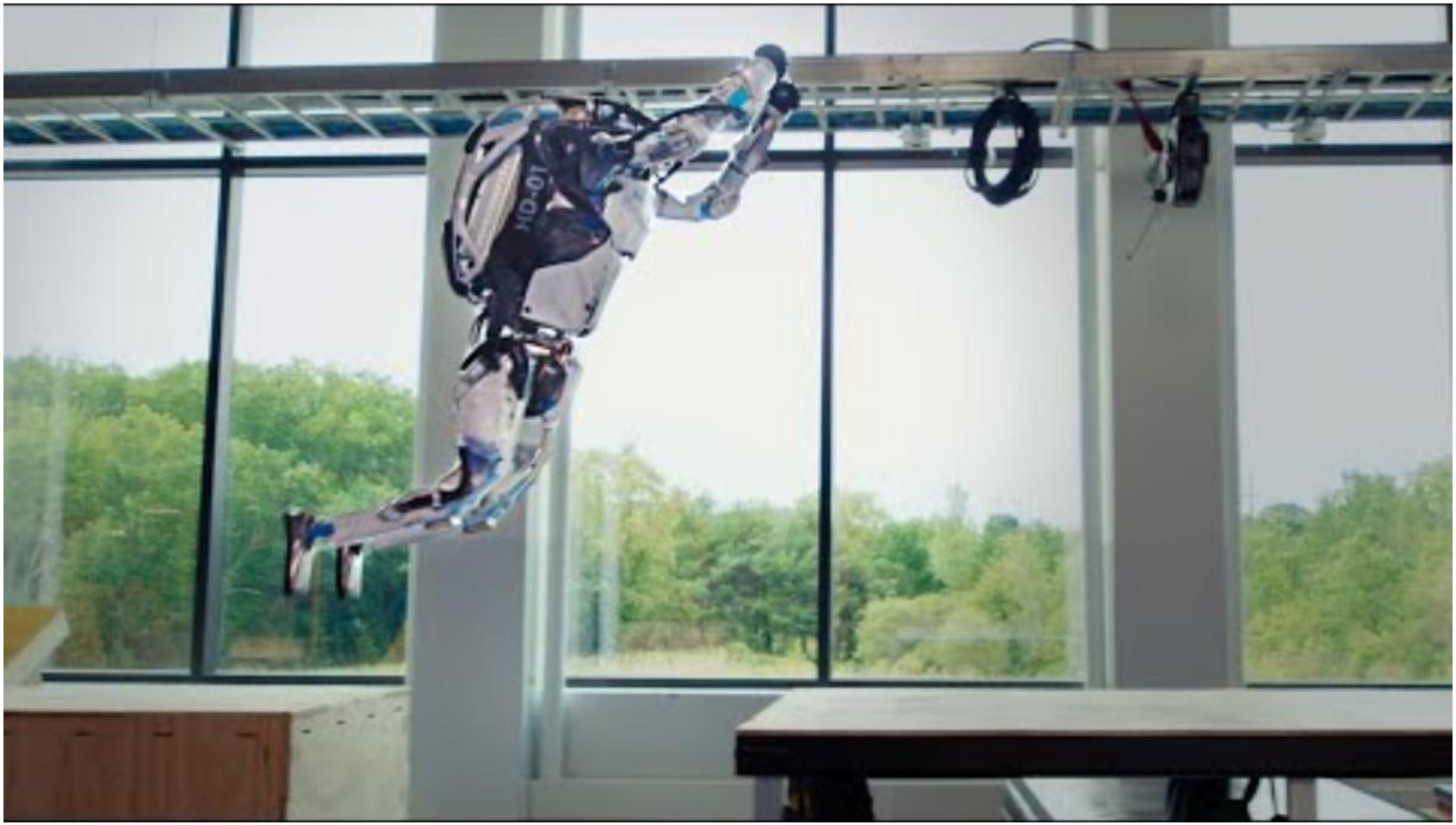
Synthesis and stabilization of complex behaviors with online trajectory optimization

Yuval Tassa, Tom Erez and Emo Todorov

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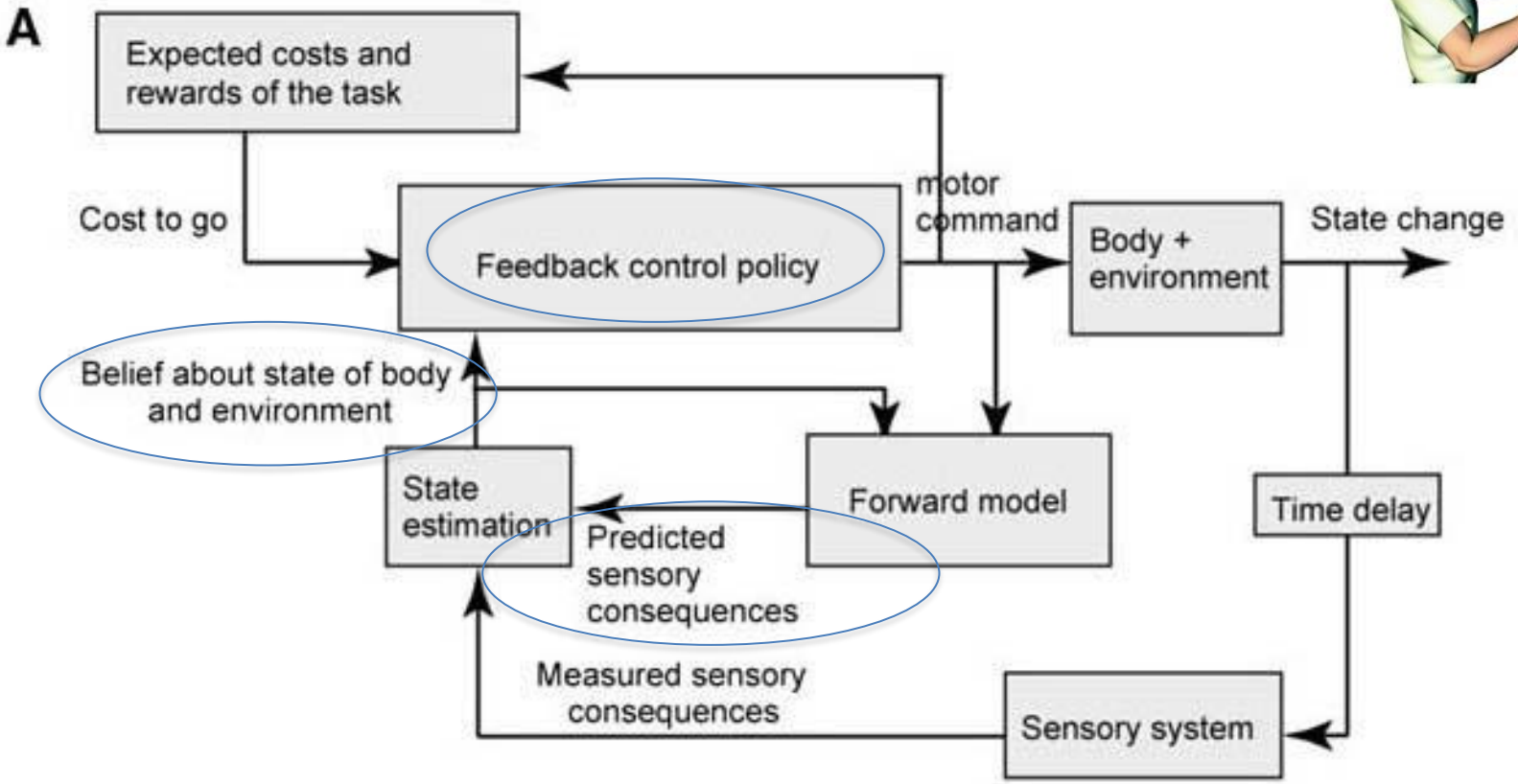
IROS 2012




HD-01

Speculating about Impairments





Krakauer and Shadmehr 2007

Suggested neural basis

